

yahoo-finance

An Unofficial Client for Yahoo Finance
Charts

`https:`
`//github.com/cgore/yahoo-finance/`

Christopher Mark Gore
`http://www.cgore.com`
`cgore@cgore.com`

Wednesday, August 26th, AD 2026

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22 Forest Glade Court, Saint Charles, Missouri 63304 USA.

Web: <http://www.cgore.com>

Email: cgore@cgore.com

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Introduction

`yahoo-finance` is a Common Lisp client for Yahoo Finance chart data. It is *not* an official Yahoo product. Yahoo has not offered a supported public Finance API since 2017 [4]. This library talks to the unofficial JSON chart endpoint that powers `https://finance.yahoo.com`:

```
https://query2.finance.yahoo.com/v8/finance/chart/AAPL
```

It is also not Paul Nathan's archived `cl-yahoo-finance` [3], which called the old `ichart` CSV and YQL surfaces. Those URLs are gone. This library starts from scratch against v8 chart, the same way `cl-coingecko` [2] wraps CoinGecko.

The thin layer lives in `yahoo-finance/rest-api`: HTTP, a browser User-Agent (Yahoo returns 429 without one), 429 retries, and `get-chart`. The `yahoo-finance` package re-exports that layer and adds the helpers you actually want: universal times, `ohlcv-history` as a list of `ohlcv-bar` objects, a translator `ohlcv-bars-for-ingest` for candlesticks, and a meta plist (currency, exchange, instrument type).

This library does *not* depend on `candlesticks` [1]. `Candlesticks` is a PostgreSQL store. An application such as `Livermore` or `Limbic` loads both and calls `ingest-candlesticks`.

v0.1 wraps only the chart endpoint. `Quote`, `quoteSummary`, `options`, and `search` need Yahoo's cookie/crumb dance and are out of scope.

You can download the library from GitHub at:
`https://github.com/cgore/yahoo-finance`

Getting Lisp

Before using this library you need a working Lisp. I use and recommend SBCL, Steel Bank Common Lisp, which is available at:
`http://www.sbcl.org`

If you are using Debian or a similar Linux distribution (including Ubuntu), you can just run as root:
`apt-get install sbcl sbcl-doc sbcl-source`

Using the Library

Clone the repository next to your other local systems, or symlink the ASDF file into a directory on `asdf:*central-registry*` (I use `~/programming/lisp/systems/`).

```
mkdir -p ~/programming/lisp
cd ~/programming/lisp
git clone git@github.com:cgore/yahoo-finance.git
ln -s ~/programming/lisp/yahoo-finance/yahoo-finance.asd \
    ~/programming/lisp/systems/
```

The system depends on `Dexador`, `function-cache`, `Quri`, `Σ`, and `Yason`. Those come from `Quicklisp`.

```
(ql:quickload :yahoo-finance)
(in-package :yahoo-finance)

(multiple-value-bind (bars meta)
  (ohlc-history "AAPL" :range "1mo")
  (list (length bars)
        (getf meta :currency)
        (getf meta :full-exchange-name)
        (first bars)))
```

`ohlc-history` defaults to a full daily history via `period1/period2`. Do not send `range=max` with `interval=1d` to the raw endpoint: Yahoo downsamples that pair to quarterly bars. `ohlc-history` rewrites it.

Testing

Specs are `sigma/behavior` / `should` forms and run at load time. `asdf:test-system` reloads the sources so those assertions run again. They do not hit the network; a checked-in five-day AAPL fixture covers parsing.

```
(asdf:test-system :yahoo-finance)
```

To also call Yahoo, set `YAHOO_FINANCE_LIVE_TESTS=1` and run `asdf:test-system` again. The live suite uses a short window. HTTP 429 is treated as a skip.

Have fun!

Chapter 1

The yahoo-finance/rest-api Package

The thin HTTP layer. It sends a browser `User-Agent`, retries 429s, parses JSON with `Yason` (arrays as vectors), and exposes one wrapper for Yahoo's unofficial chart endpoint.

1.1 Variables

1.1.1 The `*query2*` Variable

`https://query2.finance.yahoo.com` — the default host. `Query1` 429s more readily in practice.

1.1.2 The `*query1*` Variable

`https://query1.finance.yahoo.com`. Set `*api*` to this if `query2` is unhappy.

1.1.3 The `*api*` Variable

The host actually used for requests. Defaults to `*query2*`.

1.1.4 The `*user-agent*` Variable

A Chrome-shaped `User-Agent` string. Yahoo returns 429 for requests that look like a bare HTTP library.

1.1.5 The `*api-cache-timeout-seconds*` Variable

How long `http-get-cached` and `http-get-json-cached` keep a response. Default 60.

1.1.6 The `*max-retries*` Variable

How many times to retry a 429. Default 3.

1.1.7 The `*retry-wait-seconds*` Variable

Base wait, in seconds, between 429 retries. Attempt n waits $n \times$ this value. Default 2.

1.2 Functions

1.2.1 The `canonicalize-ticker` Function

Yahoo tickers are uppercase strings: AAPL, BRK-B, ^GSPC. Symbols are upcased too.

Syntax

```
(canonicalize-ticker symbol)
```

1.2.2 The `get-chart` Function

Raw GET `/v8/finance/chart/symbol`. Returns the top-level Yason hash table. This is the thin wrapper: it does *not* rewrite `range=max`. Use `ohlcv-history` for a Lisp series that avoids Yahoo's quarterly down-sample.

Syntax

```
(get-chart symbol &key interval range period1 period2 events
  include-adjusted-close include-pre-post cache)
  include-adjusted-close defaults to true. cache defaults to true.
  nil query values are omitted.
```

1.2.3 The `chart-path` Function

The request path, with the ticker URL-encoded so ^GSPC works.

1.2.4 The `chart-uri` Function

The full URL `get-chart` would request. Useful as a candlesticks retrieval endpoint.

1.2.5 The `unwrap-chart` Function

The first `chart.result` element of a payload, or a `chart-error` if Yahoo set `chart.error` or the result is empty. Yahoo sometimes returns HTTP 200 with an error object (a delisted symbol).

Syntax

```
(unwrap-chart payload &key url)
```

1.3 Conditions

1.3.1 The `api-error` Type

An error talking to Yahoo. Accessors: `api-error-status`, `api-error-url`, `api-error-code`, `api-error-message`, `api-error-timestamp`, `api-error-body`.

1.3.2 The `rate-limited` Type

HTTP 429. Subclass of `api-error`. `http-get` retries these with backoff before signalling.

1.3.3 The `chart-error` Type

The chart payload carried an error object, or had no result. Accessors `chart-error-code` and `chart-error-description` hold Yahoo's `code` / `description` fields.

The package also exports the shared HTTP helpers (`http-get`, `http-get-json`, `api-get`, `query builders`, `json-ref`) in the same style as `coingecko/rest-api`.

Chapter 2

The yahoo-finance Package

The package you use from application code. It re-exports every external symbol of `yahoo-finance/rest-api` and adds time conversion, window normalisation, and `ohlc-history`.

2.1 Time

Yahoo timestamps are UNIX seconds. Common Lisp wants universal time.

2.1.1 The `unix-epoch` Parameter

Universal time of the UNIX epoch, 1970-01-01T00:00:00Z.

2.1.2 The `unix->universal` Function

Convert a UNIX timestamp in seconds or milliseconds to universal time. Values above 10^{11} are treated as milliseconds.

2.1.3 The `universal->unix` Function

The inverse: universal time to UNIX seconds.

2.1.4 The `canonicalize-unix` Function

Turn a value into a UNIX-seconds string for `period1/period2`. Integers greater than 10^{11} are milliseconds; integers greater than $3 \cdot 10^9$ are universal times; everything else is already unix seconds.

2.2 Windows

2.2.1 The canonicalize-interval Function

Map a candlesticks short name or a Yahoo interval onto Yahoo's strings. "d" → 1d, "w" → 1wk, "M" → 1mo, "m" → 1m, "h" → 1h. The Lisp reader upcases, so :m and :M are the same symbol and mean minute; pass the string "M" or :mo for a month.

2.2.2 The normalize-chart-window Function

Return a plist :interval :range :period1 :period2 suitable for get-chart.

:from/:to (or period1/period2) win over range. range=max with a daily-or-longer interval is rewritten to period1=0 / period2=now so Yahoo does not downsample to quarterly bars. With no window at all, daily-or-longer defaults to that full-history pair.

Syntax

```
(normalize-chart-window &key interval range from to period1
period2)
```

2.3 Bars

2.3.1 The chart Function

Alias for get-chart: the raw Yason payload.

2.3.2 The chart-result Function

Unwrapped chart.result[0] after normalize-chart-window. Signals chart-error on an empty or error payload.

2.3.3 The chart-meta Function

A plist of what Yahoo said about a result: :symbol, :currency, :exchange-name, :full-exchange-name, :instrument-type, :timezone, :exchange-timezone-name, :gmt-offset, :first-trade-date (universal time), :data-granularity, :long-name, :short-name, and optional :endpoint. Mapping NasdaqGS onto a candlesticks exchange slug is the caller's job.

2.3.4 The ohlc-bar Type

One Yahoo chart bar: as-traded open, high, low, close, optional volume, and optional adjusted close. Time is a universal time. Accessors are `ohlc-bar-time`, `ohlc-bar-open`, `ohlc-bar-high`, `ohlc-bar-low`, `ohlc-bar-close`, `ohlc-bar-volume`, and `ohlc-bar-adjusted-close`. `ohlc-bar-p` is true of instances. `make-ohlc-bar` builds one from keyword arguments.

This is not a `candlesticks:candlestick`: there are no instruments, duration, or retrieval.

2.3.5 The result->bars Function

A list of `ohlc-bar` from a chart result. Bars whose OHLC are all null are dropped.

2.3.6 The ohlc-bar-tuple Function

One ingest row for `candlesticks:ingest-candlesticks`: (time open high low close volume adjusted-close).

2.3.7 The ohlc-bars-for-ingest Function

Map a sequence of `ohlc-bar` through `ohlc-bar-tuple`. This library does not depend on `candlesticks`; the caller passes the result into `ingest-candlesticks`.

2.3.8 The ohlc-history Function

The main entry point. Returns two values: a list of `ohlc-bar`, and a chart-meta plist (including the request `:endpoint`). Default window is a full daily history via `period1/period2`, not `range=max`.

Syntax

(ohlc-history symbol &key interval range from to period1 period2 events include-adjusted-close include-pre-post cache)

Examples

```
(ohlc-history "AAPL")
(ohlc-history "AAPL" :range "1y")
(ohlc-history :msft :interval "w" :range "5y")
(ohlc-history "AAPL"
  :from (encode-universal-time 0 0 0 1 1 2020 0)
  :to   (encode-universal-time 0 0 0 1 1 2024 0))
```

2.3.9 The load-chart-fixture Function

Parse a checked-in chart JSON fixture under `source/fixtures/`. Used by load-time specs; default name `aapl-chart-5d.json`.

2.4 Version

`version-string`, `version-list`, and the three `version-*` parameters are re-exported from `yahoo-finance/system`. This is version 0.1.0.

Bibliography

- [1] GORE, C. M. `candlesticks`. <https://github.com/cgore/candlesticks>, 2026.
- [2] GORE, C. M. `cl-coingecko`. <https://github.com/cgore/cl-coingecko>, 2026.
- [3] NATHAN, P. `cl-yahoo-finance`. <https://github.com/pnathan/cl-yahoo-finance>, 2025. Archived. Talked to `ichart CSV` and `YQL`, both discontinued.
- [4] YAHOO. Yahoo finance API discontinuation. <https://developer.yahoo.com/>, 2017.